

Definition. (Action) In this document, we consider the following integrand whose integral would give the action (notice that we are using a capital L to denote $\mathcal{L}\sqrt{-g}d^4x$ where $\mathcal{L}$ would be the Lagrangian density that we know. In this document, we call both the “lagrangian density” exchangeably):

$$L = (R - V) \star 1 + L_{KinS} + L_{KinA} + L_{CS}$$

where V , L_{KinS} , L_{KinA} , and L_{CS} are defined as

$$V = -4g^2 \sum_i (Y_i^2 + \tilde{Y}_i^2)$$

$$L_{KinS} = -\frac{1}{2} \sum_i \left((\partial\varphi_i)^2 + e^{2\varphi_i} (\partial\chi_i)^2 \right) \star 1$$

$$\begin{aligned} L_{KinA} = -\frac{1}{2}|W|^{-2} & \left[P_0 \left(\tilde{Y}_1^2 \tilde{Y}_2^2 \tilde{Y}_3^2 \star F_1 \wedge F_1 + \tilde{Y}_1^2 Y_2^2 Y_3^2 \star F_2 \wedge F_2 \right. \right. \\ & \left. \left. + Y_1^2 \tilde{Y}_2^2 Y_3^2 \star F_3 \wedge F_3 + Y_1^2 Y_2^2 \tilde{Y}_3^2 \star F_4 \wedge F_4 \right) \right. \\ & \left. + 2P_1 b_2 b_3 \left(\tilde{Y}_1^2 \star F_1 \wedge F_2 - Y_1^2 \star F_3 \wedge F_4 \right) \right. \\ & \left. + 2P_2 b_1 b_3 \left(\tilde{Y}_2^2 \star F_1 \wedge F_3 - Y_2^2 \star F_2 \wedge F_4 \right) \right. \\ & \left. + 2P_3 b_1 b_2 \left(\tilde{Y}_3^2 \star F_1 \wedge F_4 - Y_3^2 \star F_2 \wedge F_3 \right) \right] \end{aligned}$$

$$\begin{aligned} L_{CS} = -|W|^{-2} & \left[b_1 b_2 b_3 \left(\tilde{Y}_1^2 \tilde{Y}_2^2 \tilde{Y}_3^2 F_1 \wedge F_1 + \tilde{Y}_1^2 Y_2^2 Y_3^2 F_2 \wedge F_2 \right. \right. \\ & \left. \left. + Y_1^2 \tilde{Y}_2^2 Y_3^2 F_3 \wedge F_3 + Y_1^2 Y_2^2 \tilde{Y}_3^2 F_4 \wedge F_4 \right) \right. \\ & \left. + b_1 (P_0 + 2b_2^2 b_3^2) \left(\tilde{Y}_1^2 F_1 \wedge F_2 - Y_1^2 F_3 \wedge F_4 \right) \right. \\ & \left. + b_2 (P_0 + 2b_1^2 b_3^2) \left(\tilde{Y}_2^2 F_1 \wedge F_3 - Y_2^2 F_2 \wedge F_4 \right) \right. \\ & \left. + b_3 (P_0 + 2b_1^2 b_2^2) \left(\tilde{Y}_3^2 F_1 \wedge F_4 - Y_3^2 F_2 \wedge F_3 \right) \right] \end{aligned}$$

Meanwhile, the implicit functions Y_i^2 , $\tilde{Y}_i^2$, and b_i are defined as

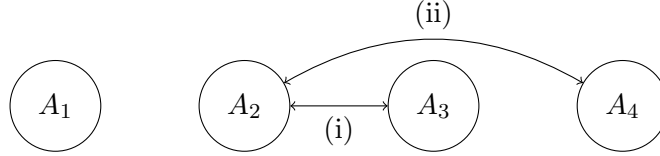
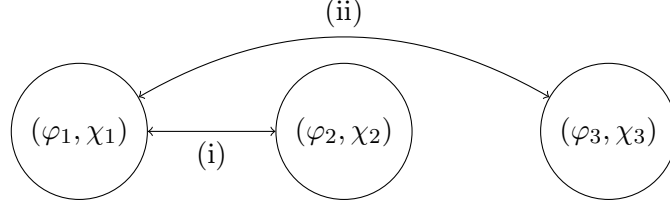
$$Y_i^2 = e^{\varphi_i} \quad \tilde{Y}_i^2 = e^{-\varphi_i} + \chi_i^2 e^{\varphi_i} \quad b_i = \chi_i e^{\varphi_i}$$

and additionally,

$$\begin{aligned} P_0 &= 1 + b_1^2 + b_2^2 + b_3^2 & W &= P_0 - 2i b_1 b_2 b_3 \\ P_1 &= 1 - b_1^2 + b_2^2 + b_3^2 & P_2 &= 1 + b_1^2 - b_2^2 + b_3^2 & P_3 &= 1 + b_1^2 + b_2^2 - b_3^2 \end{aligned}$$

Theorem. (Symmetries in the action) We now investigate the symmetries inside the action to find the number of equation of motions we need to calculate up to permutation. Assuming no symmetry, we have a total of 6 independent fields φ_i and χ_i (for $i \in \{1, 2, 3\}$) and 4 gauge fields A_j (for $j \in \{1, 2, 3, 4\}$), and one metric field $g_{\mu\nu}$, giving us a total of $6 + 4 + 1$ equations.

First notice that V and L_{KinS} are not affected by any permutation of identical fields. It therefore suffices to examine the symmetries in the L_{KinA} and L_{CS} term. By careful observation, we find the following two symmetries among fields, where permutations with the same label are acted strictly together.



This tells us that we only need to find one equation for each scalar field, two equations from the gauge fields, and one metric field, giving us a total of $1 + 2 + 1 = 5$ equations.

Definition. (The truncation limit) We define the truncation limit as the following set of limits

$$\begin{cases} \phi_1 \rightarrow \xi \\ \chi_1 \rightarrow \chi \\ A_1 \text{ and } A_2 \rightarrow A_3 \\ A_3 \text{ and } A_4 \rightarrow A_1 \end{cases}$$

While the remaining fields are put to 0. We can additionally impose the following limits such that the equations of motion would reduce to the one in Cassani's paper.

$$\begin{cases} A_i \rightarrow \frac{1}{\sqrt{2}} A_i \\ g \rightarrow \frac{1}{2} g \end{cases}$$

Proposition. (Variation with respect to the dilaton field φ_1) With the above definitions, our strategy in finding and verifying the equations of motion for the scalar fields are straightforward:

- (i) We first gather terms that are relevant in the variation process
- (ii) By simply differentiating the terms in the lagrangian density with respect to the scalar term of interest, we find $\delta L_{KinA}/\delta\varphi_1$ (the term L_{KinA} can be replaced with any term in the Lagrangian density and the field φ_1 can be replaced with any scalar field)
- (iii) We verify the equations by taking the truncation limit and the Cassani paper conventions defined above, we verify that our scalar field equations are indeed correct.

In this document, we only show the implicit versions of the equations of motion, while the entirety is displayed in the mathematica file.

$$\delta_{\varphi_1} S = \int \delta\varphi_1 \left[-4g^2(e^{\varphi_1} - e^{-\varphi_1} + \chi_1^2 e^{\varphi_1}) \star 1 + d(\star d\varphi_1) - e^{2\varphi_1} d\chi_1 \wedge \star d\chi_1 + 1 + \frac{\delta L_{KinS}}{\delta\varphi_1} + \frac{L_{KinA}}{\delta\varphi_1} + \frac{\delta L_{CS}}{\delta\varphi_1} \right]$$

We therefore have

$$0 = -4g^2(e^{\varphi_1} - e^{-\varphi_1} + \chi_1^2 e^{\varphi_1}) \star 1 + d(\star d\varphi_1) - e^{2\varphi_1} d\chi_1 \wedge \star d\chi_1 + \frac{\delta L_{KinS}}{\delta\varphi_1} + \frac{\delta L_{KinA}}{\delta\varphi_1} + \frac{\delta L_{CS}}{\delta\varphi_1}$$

Proposition. (Variation with respect to the axion field χ) Again, we have the implicit version as

$$\delta_{\chi_1} S = \int \delta\chi_1 \left[-8g^2\chi_1 e^{\varphi_1} \star 1 - \frac{1}{2}d(e^{2\varphi_1} \star d\chi_1) + \frac{\delta L_{KinS}}{\delta\chi_1} + \frac{\delta L_{KinA}}{\delta\chi_1} + \frac{\delta L_{CS}}{\delta\chi_1} \right]$$

We therefore have

$$0 = -8g^2\chi_1 e^{\varphi_1} \star 1 - \frac{1}{2}d(e^{2\varphi_1} \star d\chi_1) + \frac{\delta L_{KinS}}{\delta\chi_1} + \frac{\delta L_{KinA}}{\delta\chi_1} + \frac{\delta L_{CS}}{\delta\chi_1}$$

Proposition. (Variation with respect to gauge field A_1) For the gauge fields, we perform a straight forward variation and find

$$\delta_{A_1} S = \int \delta_{A_1} (L_{KinS} + L_{KinA} + L_{CS})$$

Resulting in

$$0 = d\left(|W|^{-2}\left[-\frac{1}{2}P_0\tilde{Y}_1^2\tilde{Y}_2^2\tilde{Y}_3^2\star F_1\right.\right. \\ \left.-P_1b_2b_3\tilde{Y}_1^2\star F_2 - P_2b_1b_3\tilde{Y}_2^2\star F_3 - P_3b_1b_2\tilde{Y}_3^2\star F_4\right. \\ \left.-b_1b_2b_3\tilde{Y}_1^2\tilde{Y}_2^2\tilde{Y}_3^2\star F_1\right. \\ \left.-b_1(P_0+2b_2^2b_3^2)\tilde{Y}_1^2\star F_2 - b_2(P_0+2b_1^2b_3^2)\tilde{Y}_2^2\star F_3 + b_3(P_0+2b_1^2b_2^2)\tilde{Y}_3^2\star F_4\right]\Bigg)$$

Proposition. (Variation with respect to gauge field A_2)

$$\delta_{A_1}S = \int \delta_{A_1}(L_{KinS} + L_{KinA} + L_{CS})$$

Resulting in

$$0 = d\left(|W|^{-2}\left[P_0\tilde{Y}_1^2Y_2^2Y_3^2\star F_2 + 2P_1b_2b_3\tilde{Y}_1^2\star F_1 - 2P_2b_1b_3Y_2^2\star F_4 - 2P_3b_1b_2Y_3\star F_3\right]\right)$$

Proposition. (Variation with respect to metric field $g_{\mu\nu}$) The only terms in the lagrangian density that depend on the metric are $R\star 1$, L_{KinS} , and L_{KinA} . Before we apply the variation, we first write the relevant terms in component form.

$$R = g^{\mu\nu}R_{\mu\nu} \quad \mathcal{L}_{kinS} = -\frac{1}{2}g^{\mu\nu}\sum_i(\partial_\mu\varphi_i\partial_\nu\varphi_i + e^{2\varphi_i}\partial_\mu\chi_i\partial_\nu\chi_i)$$

$$\mathcal{L}_{KinA} = -\frac{1}{8}|W|^{-2}\left[P_0\left(\tilde{Y}_1^2\tilde{Y}_2^2\tilde{Y}_3^2(F_1)^{\mu\nu}(F_1)_{\mu\nu} + \tilde{Y}_1^2Y_2^2Y_3^2(F_2)^{\mu\nu}(F_2)_{\mu\nu}\right.\right. \\ \left.+Y_1^2\tilde{Y}_2^2Y_3^2(F_3)^{\mu\nu}(F_3)_{\mu\nu} + Y_1^2Y_2^2\tilde{Y}_3^2(F_4)^{\mu\nu}(F_4)_{\mu\nu}\right) \\ \left.+2P_1b_2b_3\left(\tilde{Y}_1^2(F_1)^{\mu\nu}(F_2)_{\mu\nu} - Y_1^2(F_3)^{\mu\nu}(F_4)_{\mu\nu}\right)\right. \\ \left.+2P_2b_1b_3\left(\tilde{Y}_2^2(F_1)^{\mu\nu}(F_3)_{\mu\nu} - Y_2^2(F_2)^{\mu\nu}(F_4)_{\mu\nu}\right)\right. \\ \left.+2P_3b_1b_2\left(\tilde{Y}_3^2(F_1)^{\mu\nu}(F_4)_{\mu\nu} - Y_3^2(F_2)^{\mu\nu}(F_3)_{\mu\nu}\right)\right]$$

where we have omitted the volume element $\sqrt{-g}d^4x$. We know that the Einstein field equation is given as

$$\frac{\delta\mathcal{L}}{\delta g^{\mu\nu}} - \frac{1}{2}g_{\mu\nu}\mathcal{L} = 0$$

which we can compute for our action to be

$$\begin{aligned}
R_{\mu\nu} - \frac{1}{2} \sum_i (\partial_\mu \varphi_i \partial_\nu \varphi_i + e^{2\varphi_i} \partial_\mu \chi_i \partial_\nu \chi_i) \\
- \frac{1}{4} |W|^{-2} \left[P_0 \left(\tilde{Y}_1^2 \tilde{Y}_2^2 \tilde{Y}_3^2 (F_1)_\mu{}^\rho (F_1)_{\nu\rho} + \tilde{Y}_1^2 Y_2^2 Y_3^2 (F_2)_\mu{}^\rho (F_2)_{\nu\rho} \right. \right. \\
\quad \left. \left. + Y_1^2 \tilde{Y}_2^2 Y_3^2 (F_3)_\mu{}^\rho (F_3)_{\nu\rho} + Y_1^2 Y_2^2 \tilde{Y}_3^2 (F_4)_\mu{}^\rho (F_4)_{\nu\rho} \right) \right. \\
\quad \left. + 2P_1 b_2 b_3 \left(\tilde{Y}_1^2 (F_1)_\mu{}^\rho (F_2)_{\nu\rho} - Y_1^2 (F_3)_\mu{}^\rho (F_4)_{\nu\rho} \right) \right. \\
\quad \left. + 2P_2 b_1 b_3 \left(\tilde{Y}_2^2 (F_1)_\mu{}^\rho (F_3)_{\nu\rho} - Y_2^2 (F_2)_\mu{}^\rho (F_4)_{\nu\rho} \right) \right. \\
\quad \left. + 2P_3 b_1 b_2 \left(\tilde{Y}_3^2 (F_1)_\mu{}^\rho (F_4)_{\nu\rho} - Y_3^2 (F_2)_\mu{}^\rho (F_3)_{\nu\rho} \right) \right] \\
- \frac{1}{2} g_{\mu\nu} (R - V + L_{KinS} + L_{KinA} + L_{CS})
\end{aligned}$$

where we have wrote the last term implicitly.